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Bracketing Inference with Uncertainty Quantification: A Reliability Pipeline for Neural Aerodynamic Surrogates

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Abstract

Data-driven surrogates for computational fluid dynamics (CFD) offer dramatic acceleration of the automotive design cycle, but their adoption requires assessing when a prediction can be trusted. As surrogate accuracy on public benchmarks approaches saturation, the barrier to deployment shifts from performance to reliability: a trained model never refuses to predict, and silent extrapolation on novel geometries poses a direct risk to engineering decisions. This study proposes a two-stage reliability pipeline that brackets inference with uncertainty quantification (UQ): a pre-inference out-of-distribution (OOD) detection step screens whether a geometry lies within the model’s domain of competence, and a post-inference confidence interval step quantifies the spatial distribution of prediction uncertainty. The pipeline is validated on a transformer-based surrogate trained on 511 aerodynamic simulations across five vehicle types. For OOD detection, Mahalanobis distance in the model’s learned feature space provides clear separation of unseen vehicle types from in-distribution data, with higher scores correlating with larger prediction errors. For post-inference diagnostics, Monte Carlo Dropout produces spatially resolved uncertainty maps whose high-variance regions coincide with areas of large prediction error, though its confidence intervals are miscalibrated. The specific methods are not unique solutions; the contribution lies in demonstrating that pre- and post-inference UQ can be assembled into a practical safeguard for deploying neural surrogates in industrial scientific machine learning (SciML) workflows.

Keywords: uncertainty quantification, scientific machine learning, computational fluid dynamics, out-of-distribution detection, Monte Carlo Dropout, neural surrogate

Article Highlights

- Neural surrogates never refuse to predict; a two-stage UQ pipeline is the missing safeguard for engineering use.
- Pre-inference screening catches unreliable predictions before committing compute, preventing silent failures.
- Post-inference confidence intervals severely underestimate true errors, a critical caveat for practitioners.

1 Introduction

Scientific machine learning (SciML) surrogates for computational fluid dynamics (CFD) have made rapid progress in recent years. Open-source aerodynamic datasets such as DrivAerNet [1], DrivAerNet++ [2], and DrivAerStar [3], together with benchmarks built upon them such as CarBench [4], have enabled systematic evaluation of data-driven approaches. On these benchmarks, transformer-based architectures—including the Anchored-Branched Universal Physics Transformer (AB-UPT) [5], Transolver [6], GeoTransolver [7], and Multi-Scale Patch Transformer (MSPT) [8]—have demonstrated strong predictive accuracy for three-dimensional aerodynamic fields. With benchmark performance approaching a plateau, the research frontier is shifting from *model accuracy* toward *deployment readiness*: the question is no longer whether a neural surrogate can predict aerodynamic fields, but under what conditions its predictions can be trusted for engineering decisions.

This shift is critical because a trained neural surrogate will *never refuse to predict*. Given any input geometry, the model produces a plausible-looking pressure field and force coefficient—regardless of whether that geometry resembles anything in the training set. In conventional CFD, unreliable results manifest as divergence, poor residual convergence, or unphysical flow features that experienced engineers can identify. Machine learning surrogates offer no such warning signals: predictions on out-of-distribution (OOD) inputs are visually indistinguishable from those on in-distribution data. Without an explicit reliability mechanism, silent extrapolation failures pose a direct risk to design decisions.

Uncertainty quantification (UQ) provides two complementary safeguards in this context:

1. **Pre-inference screening:** Before committing to a prediction, determine whether the input geometry is sufficiently similar to the training data to warrant trust. A lightweight and reliable OOD detection mechanism is therefore needed before computationally intensive inference, so that computing resources are not wasted on predictions the model is ill-equipped to make.
2. **Post-inference diagnostics:** For geometries that pass the OOD screen, quantify *where* on the surface and *by how much* the prediction may be uncertain. This

requires a UQ method that provides confidence intervals in the traditional statistical sense. For practical deployment, however, such post-inference diagnostics must not impose excessive computational overhead, so as to minimize impact on the end-user experience.

A further constraint of industrial SciML, largely absent from academic benchmarks, is *data scarcity*. Each training sample is the product of a high-fidelity CFD simulation requiring hours to days of compute time on high-performance computing systems. In an industrial setting, generating thousands of simulation samples is difficult to justify given the associated cost. The dataset used in this study—511 simulations across five vehicle types—reflects this reality.

The present study addresses these industrial needs by implementing and validating two critical UQ elements within a production-oriented SciML pipeline. Rather than proposing new UQ algorithms, the contribution lies in the systematic empirical assessment of existing methods under realistic industrial constraints—limited data, proprietary geometries, transformer-scale models—and in demonstrating that a practical two-stage reliability workflow can be constructed from off-the-shelf components.

1.1 Two Sources of Uncertainty

Uncertainty in machine learning predictions is conventionally decomposed into two fundamentally distinct sources [9, 10]:

- **Aleatoric uncertainty** (data uncertainty) arises from inherent noise or variability in the data itself. In the context of CFD surrogate modeling, this includes numerical discretization artifacts, turbulence model stochasticity, and convergence tolerances in the reference simulations. Aleatoric uncertainty is *irreducible* and cannot be eliminated by collecting more training data, as it reflects fundamental limitations of the data generation process.
- **Epistemic uncertainty** (model uncertainty) arises from the model’s limited knowledge due to finite training data and constrained model capacity. This uncertainty is *reducible*: it decreases as the training set grows and the model encounters a broader diversity of geometries. Crucially, epistemic uncertainty is expected to be high for inputs that lie far from the training distribution, which is precisely the scenario of concern in the present setting.

For SciML surrogates, the practical priority is epistemic uncertainty: when a new vehicle geometry is submitted for prediction, the primary risk is not noise in the training simulations (which are high-fidelity CFD results with controlled convergence), but rather whether the model has sufficient training coverage to make reliable predictions for this particular geometry. This motivates the focus on two categories of methods: those that detect whether an input lies outside the training distribution (OOD detection), and those that quantify the prediction-level uncertainty for inputs that are deemed in-distribution (confidence estimation).

1.2 Related Work

Uncertainty quantification for deep neural networks has been the subject of extensive research. He et al. [9] present a comprehensive taxonomy of UQ methods organized by the type of uncertainty source they address. They distinguish *model uncertainty* (epistemic), arising from limited training data, sub-optimal architectures, or distribution shift between training and inference, from *data uncertainty* (aleatoric), arising from inherent noise or ambiguity in the observations. Methods for model uncertainty include Bayesian neural networks (BNNs), deep ensembles, MC Dropout [11], and distance-aware neural networks; methods for data uncertainty include parametric predictive distributions and deep generative models (VAEs, GANs, diffusion models). The survey also covers approaches that capture both sources simultaneously, such as evidential deep learning and conformal prediction. A key insight from this taxonomy is that no single method addresses all uncertainty sources: BNN and ensemble approaches primarily target epistemic uncertainty but incur high computational cost due to multiple forward passes, while MC Dropout offers a practical approximation at the expense of calibration quality. Distance-aware methods, which measure how far a test sample’s features lie from the training distribution, are particularly relevant for OOD detection, the primary concern in the present setting.

In the specific context of Scientific Machine Learning (SciML), Psaros et al. [12] provide a systematic framework covering uncertainty modeling, solution methods, evaluation metrics, and post-hoc calibration approaches for physics-informed neural networks (PINNs) and neural operators. Their work highlights that SciML introduces additional uncertainty sources beyond those in standard DL: imperfect knowledge of governing equations, stochastic boundary/initial conditions, and discretization artifacts from the underlying PDE solvers used to generate training data. Through an extensive comparative study on prototype problems, including function approximation, forward/inverse PDE problems, and operator learning, they demonstrate that different UQ methods exhibit markedly different strengths depending on the problem setting, and that no single approach dominates across all scenarios. They also emphasize the importance of proper evaluation metrics (calibration, sharpness, coverage) as standard prediction accuracy alone fails to capture UQ quality.

The present work builds on both perspectives. From the general DL taxonomy [9], two representative methods are selected for empirical evaluation: MC Dropout as a practical epistemic uncertainty estimator, and Mahalanobis distance as a distribution-aware OOD detector. These are not claimed to be the only viable choices; they are adopted because they are well-established, require no architectural modification beyond standard dropout layers, and can be applied to any pretrained model. From the SciML-specific framework [12], the emphasis on evaluation beyond point-prediction accuracy is inherited, along with the unique challenge of limited training data inherent to simulation-based datasets.

1.3 Contributions

This study provides a systematic reliability assessment of a transformer-based neural surrogate for automotive external aerodynamics, using the AB-UPT architecture [5]

as a representative platform. The assessment is centered on two questions of practical engineering importance:

1. **Can we detect when the surrogate will fail?** A pre-inference OOD detection mechanism is needed so that novel geometries can be flagged before committing to a full prediction. In this study, Mahalanobis distance [13] in the model’s learned geometry-encoder feature space is shown to provide unambiguous separation of unseen vehicle types from in-distribution data with zero overlap. Furthermore, higher distance scores correlate with larger force-prediction errors, making the metric a continuous reliability indicator beyond binary OOD classification.
2. **Can we quantify *where* on the surface the surrogate is uncertain?** A post-inference confidence interval is needed to give engineers actionable information about the spatial distribution of prediction uncertainty. In this study, MC Dropout [11] is evaluated for this role: it produces spatially resolved uncertainty maps that coincide with regions of high prediction error, providing localized diagnostic information. However, its absolute uncertainty scale is shown to be severely miscalibrated, with the miscalibration worsening on OOD geometries [14].

These findings are validated on 511 production-vehicle CFD simulations spanning five vehicle types, through progressive training-set expansion experiments (1 to 5 vehicle types) and cross-architecture comparison (full model vs. surface-only variant). The results motivate a practical two-stage deployment workflow: pre-inference OOD screening followed by optional post-inference confidence estimation, applicable to any SciML surrogate regardless of the specific methods chosen for each stage.

2 Problem Description

2.1 Vehicle Types and Design Variations

The scope of this research is to develop and evaluate uncertainty quantification methods as safeguards for engineering decision-making with neural surrogates, specifically to answer two questions: (1) Is the trained model suitable for a new, previously unseen design? and (2) Which regions of the predicted flow field are most uncertain, and how much confidence should be placed in the integral metrics derived from them, such as drag or lift?

Answering these questions requires a *heterogeneous* dataset that captures the diversity of geometries encountered in real-world automotive design. OOD detection methods must be evaluated on their ability to distinguish between in-distribution (ID) and out-of-distribution (OOD) samples, and post-inference UQ methods must be assessed on how well their uncertainty estimates correlate with true prediction errors across a range of geometries.

While open-source aerodynamic datasets such as DrivAerNet [1] and DrivAerNet++ [2] have advanced data-driven CFD research, they are composed exclusively of sedan-type geometries. Cross-category generalization among three sedan sub-categories (estateback, notchback, fastback) has been reported, with good performance from a model trained on the fastback variant alone [4]. However, evaluating OOD

detection requires a truly *heterogeneous* multi-vehicle-type dataset spanning distinct body styles—midsize SUVs, crossovers, and large SUVs—where inter-class geometric differences are far greater than those among sedan variants. No publicly available dataset currently satisfies this requirement. Consequently, the authors leverage a proprietary in-house dataset comprising five different vehicle types, each with approximately 100 design variations, for a total of 511 CFD simulations. The vehicle types and dataset composition are summarized in Table 1.

Table 1: Dataset composition by vehicle type.

Vehicle Type	Number of Variants
Midsize SUV	102
Small Crossover	103
Large SUV #1	103
Large SUV #2	100
Large SUV #3	103
Total	511

Each design variant represents a unique geometric configuration of the same vehicle platform, capturing variations in body panels, fascia, spoilers, underbody, and other exterior components. These variants span a realistic range of design exploration typically encountered in the vehicle development process.

Training data were generated using PowerFLOW [15], a commercial CFD package based on the Lattice Boltzmann method [16]. Details of the simulation setup and data generation process can be found in [17].

The Large SUV #3 vehicle is notable as a pre-production design study that did not proceed to manufacturing, making it a natural test case for OOD detection: its geometry characteristics differ from the four production vehicles in the dataset. However, as shown in Figure 1, Large SUV #3 is *not* an outlier in terms of conventional aerodynamic summary statistics. Its C_d and frontal area distributions overlap substantially with those of the production vehicles, particularly Midsize SUV and Large SUV #2, meaning that simple scalar metrics cannot distinguish Large SUV #3 from the training set. This observation motivates the use of feature-space methods (Section 5.1) that capture the full geometric complexity rather than relying on low-dimensional summary statistics.

Each simulation provides:

- **Surface fields:** Static pressure and tangential force vector on the vehicle surface mesh. These are used for aerodynamic force calculation (drag, lift, side force) via surface integration.
- **Volume fields:** Static pressure and velocity vector in the three-dimensional flow domain surrounding the vehicle.

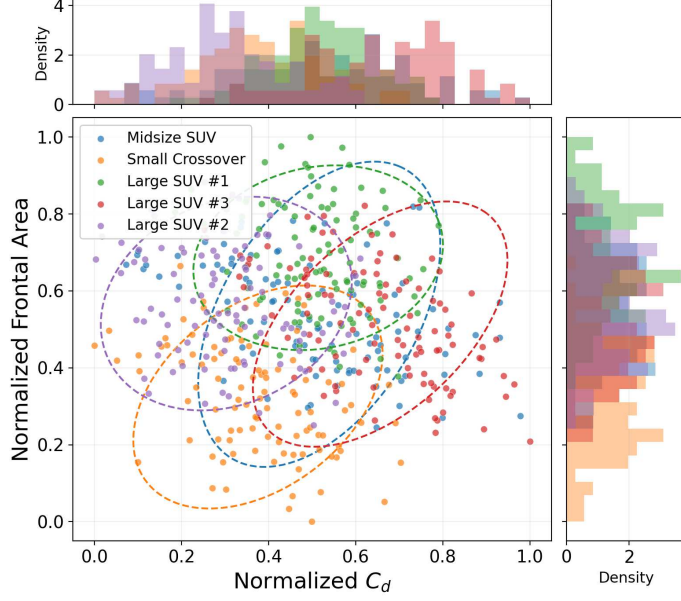


Fig. 1: Joint distribution of normalized drag coefficient and normalized frontal area across the 511 design variants, colored by vehicle type. Both axes are min–max scaled to $[0, 1]$. Marginal density plots are shown along each axis. Dashed ellipses indicate the 2σ covariance ellipse per vehicle. Despite distinct vehicle identities, the distributions overlap significantly, demonstrating that C_d and area alone are insufficient to identify OOD geometries.

2.2 Data Splits

For each experiment configuration, the data is randomly split into training (80%), validation (10%), and test (10%) sets. When fewer than five vehicle types are used for training, the excluded vehicle types’ samples are classified as OOD during evaluation, providing a natural testbed for OOD detection.

3 Neural Surrogate Architecture

3.1 Anchored-Branched Universal Physics Transformer (AB-UPT)

The AB-UPT [5] is a multi-branch Transformer neural operator designed for large-scale physics simulations. AB-UPT extends the Universal Physics Transformer (UPT) [18] with two key innovations: (i) a multi-branch architecture that separates geometry encoding, surface prediction, and volume prediction into distinct interacting branches, and (ii) anchor attention that enables scalability to industry-scale meshes with millions of points.

3.1.1 Architecture Overview

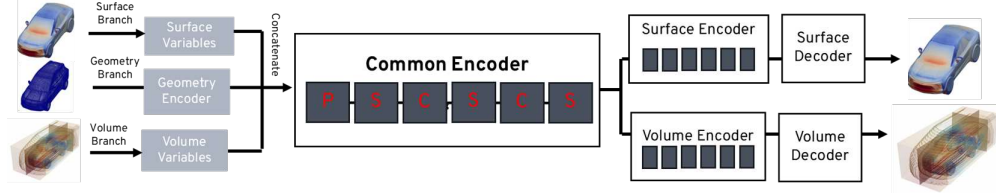


Fig. 2: Overview of the AB-UPT architecture [5]. The model takes a 3D geometry as input and encodes it through three branches: surface variables, geometry encoding, and volume variables. The concatenated representations pass through a common encoder composed of configurable physics blocks (P: Perceiver, S: Split self-attention, C: Cross-attention), followed by branch-specific surface and volume encoders and decoders that produce the final field predictions.

The model consists of three branches:

- **Geometry branch:** Encodes the input vehicle geometry (represented as a 3D point cloud) into a compact set of supernode representations through positional embedding and supernode pooling, followed by self-attention:

$$\begin{aligned} \mathbf{X}^{\text{geo}} \in \mathbb{R}^{N^{\text{geo}} \times 3} &\xrightarrow{\text{Embed} + \text{Pool}} \mathbf{S} \in \mathbb{R}^{S \times d} \\ &\xrightarrow{\text{Attention}} \mathbf{Z}^{\text{geo}} \in \mathbb{R}^{S \times d} \end{aligned} \quad (1)$$

where N^{geo} sampled geometry points are reduced to S supernodes, each with d latent dimensions.

- **Surface branch:** Predicts surface fields (pressure, tangential force) at M^{sur} anchor points through cross-attention with geometry features and self-attention among anchors.
- **Volume branch:** Predicts volumetric fields (pressure, velocity) at M^{vol} anchor points using the same mechanism as the surface branch, with additional cross-attention to exchange information between surface and volume branches.

A distinctive strength of AB-UPT is that both the supernodes and anchor points are *randomly sampled* at each training epoch. This stochastic subsampling serves a dual purpose. First, it implements an effective downsampling strategy that reduces the token count presented to the Transformer, mitigating the memory explosion that would otherwise arise when applying attention-based architectures to CAE meshes with millions of elements. Second, because the model observes a different subset of points at every epoch, the random sampling acts as a form of implicit data augmentation, improving generalization without requiring explicit augmentation of the training set.

The physics blocks (also referred to as *common blocks*) follow a configurable pattern. Three block types are available: Perceiver blocks (**p**) perform cross-attention from surface/volume anchors to geometry supernodes, injecting geometric context into the prediction branches. Split self-attention blocks (**s**) apply independent self-attention within each branch (surface and volume separately), allowing each to develop specialized representations. Cross-attention blocks (**c**) perform cross-attention between the surface and volume branches, enabling information exchange across prediction domains.

These common blocks share parameters across the surface and volume branches. Following the common blocks, each prediction branch has its own *mode-specific encoder blocks*: additional Transformer layers that operate exclusively on one branch and do not share parameters. This two-stage design allows the shared common blocks to learn cross-branch representations, while the mode-specific blocks specialize each branch for its respective prediction task.

3.1.2 Inference: From Anchors to Query Points

At inference time, the model predicts physical quantities at the anchor point locations. To obtain predictions at arbitrary query positions, the query coordinates are processed in chunks through cross-attention with the anchor representations. This achieves linear complexity in the number of query points, enabling prediction on meshes of arbitrary resolution.

It is important to note that inference does not require the original CFD mesh. Predictions can equally be made on a decimated mesh or a downsampled point cloud with far fewer vertices. However, mesh decimation risks deteriorating fine geometric details that influence the predicted fields, and point-cloud-based inference lacks the mesh connectivity information required for post-processing operations such as surface integration of forces or streamline generation. For these reasons, the present study performs inference directly on the full CFD mesh. Nonetheless, a CFD-level fine mesh is not a prerequisite for AB-UPT inference.

3.2 Model Configuration

The original AB-UPT publication [5] uses a latent dimension of $d = 192$, 4 attention heads, a single geometry encoder block, and 6 mode-specific encoder blocks per branch. The remaining architectural choices (block types, supernode/anchor counts, supernode radius) are identical to the values listed below.

In the present study, these hyperparameters have been re-calibrated for the target vehicle geometries, which exhibit larger spatial extent and finer surface detail than the original benchmark datasets. Table 2 summarizes both the original and the present configurations. Key differences are a wider latent dimension ($d = 512$), 8 attention heads, 6 geometry encoder blocks, and 12 mode-specific encoder blocks per branch. The common block pattern **pspsps** alternates between Perceiver and Split self-attention over 6 blocks; the cross-attention block (**c**) is available but not used in the current configuration. The resulting model has approximately 87M parameters.

Table 2: AB-UPT model configuration: original publication vs. the present study.

Parameter	Original [5]	This study
Mode	Surface + Volume	Surface + Volume [†]
Latent dimension (d)	192	512
Geometry encoder depth	1 block	6 blocks
Attention heads	4	8
Common block pattern	p s c s c s (6 blocks)	p s p s p s (6 blocks)
Mode-specific encoder blocks	6 per branch	12 per branch
Geometry sample points	65,536	65,536
Supernodes	16,384	16,384
Anchor points (surface)	16,384	16,384
Anchor points (volume)	16,384	16,384
Supernode radius	0.25	0.25

Training was performed on $8\times$ NVIDIA H100 GPUs with mixed precision (bfloat16), a learning rate of 6×10^{-5} with cosine annealing, and a batch size of 1 per GPU for 500 epochs.

[†]The MC Dropout experiments (Section 5.4) use a surface-only variant of this architecture, which omits the volume branch and cross-attention blocks. The block pattern reduces to **p** (Perceiver-only), with 12 mode-specific encoder blocks. All other hyperparameters remain identical.

4 Uncertainty Quantification Methods

Two complementary UQ methods are applied, each operating at a different level of granularity and computational cost. Following the pipeline order—pre-inference screening before post-inference diagnostics—Mahalanobis distance is presented first, followed by MC Dropout.

4.1 Mahalanobis Distance

The Mahalanobis distance [13] provides a principled measure of how far a new sample lies from the training distribution in the model’s learned feature space. Unlike MC Dropout, which quantifies prediction-level uncertainty, Mahalanobis distance operates at the geometry level, answering the question: *is this geometry similar to what the model was trained on?*

4.1.1 Feature Extraction

Features are extracted from the last layer of the geometry encoder branch, rather than from the downstream physics branches (surface or volume decoders). This choice is deliberate: the geometry encoder processes only the input shape through supernode pooling and self-attention, producing a representation of what the geometry “looks like” to the model, independent of any particular physical field being predicted. Since

the goal is to determine whether a new geometry is explainable by the model’s training experience, the geometry encoder’s feature space is the most natural domain for measuring distributional similarity. Features from the physics branches, by contrast, would conflate geometric novelty with field-prediction behavior, making it harder to isolate the question of whether the input itself is in-distribution.

Given an input geometry, a forward hook captures the output of the final geometry Transformer block:

$$\mathbf{Z}^{\text{geo}} \in \mathbb{R}^{S \times d} \xrightarrow{\text{mean pool}} \mathbf{f} \in \mathbb{R}^d \quad (2)$$

where S supernodes are mean-pooled into a single d -dimensional feature vector $\mathbf{f}$.

Rationale for mean pooling.

The supernode set $\{1, \dots, S\}$ is not fixed across samples or even across epochs: during training, S points are randomly sub-sampled from the full point cloud at each iteration, providing an inherent data augmentation that enriches the model’s geometric representation. However, this stochastic sampling means that supernode i in one sample does not correspond to the same spatial location as supernode i in another sample. Computing per-node Mahalanobis distances (i.e., treating each row of $\mathbf{Z}^{\text{geo}}$ independently) is therefore meaningless, as there is no consistent spatial correspondence across samples to anchor the comparison. Mean pooling over the supernode dimension collapses the spatially unregistered $S \times d$ tensor into a single d -dimensional summary vector $\mathbf{f}$ that serves as a global geometric fingerprint, enabling meaningful cross-sample comparison in a shared feature space.

4.1.2 Training Distribution Statistics

From the training set $\{\mathbf{f}_i\}_{i=1}^N$, the following statistics are computed:

$$\boldsymbol{\mu} = \frac{1}{N} \sum_{i=1}^N \mathbf{f}_i \quad (3)$$

$$\boldsymbol{\Sigma} = \frac{1}{N-1} \sum_{i=1}^N (\mathbf{f}_i - \boldsymbol{\mu})(\mathbf{f}_i - \boldsymbol{\mu})^\top \quad (4)$$

where $\boldsymbol{\mu}$ is the mean feature vector and $\boldsymbol{\Sigma}$ is the sample covariance matrix.

Covariance regularization.

When the number of training samples N is comparable to or smaller than the feature dimension d (as is common in SciML applications where each sample is an expensive simulation), the sample covariance matrix is rank-deficient or ill-conditioned. We employ Ledoit–Wolf shrinkage [19]:

$$\hat{\boldsymbol{\Sigma}} = (1 - \alpha) \boldsymbol{\Sigma} + \alpha \frac{\text{tr}(\boldsymbol{\Sigma})}{d} \mathbf{I} \quad (5)$$

where $\alpha \in [0, 1]$ is the shrinkage coefficient estimated automatically via the Ledoit–Wolf formula, $\text{tr}(\mathbf{\Sigma})/d$ is the average variance, and $\mathbf{I}$ is the identity matrix. This produces a well-conditioned precision matrix $\hat{\mathbf{\Sigma}}^{-1}$.

4.1.3 Distance Computation

For a new geometry with feature vector $\mathbf{f}^*$, the Mahalanobis distance is:

$$D_M(\mathbf{f}^*) = \sqrt{(\mathbf{f}^* - \boldsymbol{\mu})^\top \hat{\mathbf{\Sigma}}^{-1} (\mathbf{f}^* - \boldsymbol{\mu})} \quad (6)$$

Under the assumption that training features follow a multivariate Gaussian $\mathcal{N}(\boldsymbol{\mu}, \hat{\mathbf{\Sigma}})$, the squared Mahalanobis distance D_M^2 follows a $\chi^2(d)$ distribution, providing a natural reference for calibration: $\mathbb{E}[D_M] \approx \sqrt{d}$.

4.1.4 Implementation

A critical prerequisite for Mahalanobis distance is the invertibility of the covariance matrix $\mathbf{\Sigma} \in \mathbb{R}^{d \times d}$. The sample covariance has rank at most $\min(N-1, d)$; when $N \leq d$, the matrix is singular and $\mathbf{\Sigma}^{-1}$ does not exist. Even when N is modestly larger than d , the sample covariance is highly unstable: small perturbations in the data produce large changes in the estimated precision matrix.

In the present setting, the feature dimension is $d = 512$ while the largest training set contains $N = 408$ samples (5-vehicle model), yielding a ratio $N/d \approx 0.80 < 1$. The smaller training configurations are even more constrained: the 1-vehicle model has $N/d = 91/512 \approx 0.18$. Naïve inversion of the sample covariance is therefore impossible for any of these experiments.

This issue is addressed through **Ledoit–Wolf shrinkage** [19], which replaces the sample covariance with a convex combination of $\mathbf{\Sigma}$ and a scaled identity matrix (Section 4.1, “Covariance regularization”). The shrinkage coefficient α is determined analytically by minimizing the expected Frobenius-norm loss $\mathbb{E}\|\hat{\mathbf{\Sigma}} - \mathbf{\Sigma}_{\text{true}}\|_F^2$, requiring no cross-validation. In the implementation, `sklearn.covariance.LedoitWolf` from scikit-learn is used, which computes α and the regularized precision matrix in a single call.

The resulting α values provide a useful diagnostic:

- $\alpha \rightarrow 0$: the sample covariance is reliable ($N \gg d$); minimal shrinkage applied.
- $\alpha \rightarrow 1$: the sample covariance is unreliable ($N \ll d$); features are treated as approximately independent, and D_M approaches a scaled Euclidean distance.

In the present experiments, α ranges from approximately 0.05 (5-vehicle model, $N = 408$) to 0.12 (LoRA model, $N = 20$), reflecting the varying degrees of sample deficiency.

Feature extraction pipeline.

Feature extraction is performed via a PyTorch `register_forward_hook` attached to the last geometry Transformer block (`geometry_blocks[-1]`). During a standard forward pass, the hook intercepts the block’s output tensor $\mathbf{Z}^{\text{geo}} \in \mathbb{R}^{B \times S \times d}$ and applies

mean pooling over the supernode dimension, producing a single feature vector $\mathbf{f} \in \mathbb{R}^d$ per geometry. This approach requires no modification to the model’s computation graph or weights.

A key numerical consideration is precision: the model’s forward pass uses mixed-precision inference (`torch.autocast` with `bfloat16`), but the hook explicitly casts the captured output to `float32` before aggregation. This ensures that the downstream covariance and precision matrix computations—which are sensitive to numerical precision—operate in full 32-bit floating point.

Numerical robustness.

Before computing the sample covariance, features are checked for NaN and Inf values (which can arise from numerical instabilities in mixed-precision inference), and any affected samples are removed. Additionally, since finite-precision arithmetic can produce very small negative values in the quadratic form $(\mathbf{f}^* - \boldsymbol{\mu})^\top \hat{\boldsymbol{\Sigma}}^{-1} (\mathbf{f}^* - \boldsymbol{\mu})$, we clamp $D_M^2 \geq 0$ before taking the square root. The condition number of the covariance matrix is logged both before and after Ledoit–Wolf shrinkage as a diagnostic: a drop of several orders of magnitude (e.g., from 10^{15} to 10^3) confirms that the regularization is effective.

Two-stage evaluation pipeline.

The Mahalanobis distance evaluation proceeds in two stages:

1. **Statistics computation** : One forward pass per training sample extracts $\{\mathbf{f}_i\}_{i=1}^N$, computes $\boldsymbol{\mu}$ and $\hat{\boldsymbol{\Sigma}}^{-1}$ via Ledoit–Wolf, evaluates D_M for training samples as a reference distribution, and saves all statistics to a `.pt` file.
2. **Batch evaluation** : Loads the precomputed statistics, extracts features for all samples (including unseen geometries), computes D_M scores, and generates per-sample CSV outputs with per-vehicle breakdowns for analysis.

Each geometry is processed individually (batch size 1) since point cloud sizes vary across geometries. Feature extraction is parallelized across available GPUs via round-robin assignment of model replicas.

4.2 MC Dropout

MC Dropout [11] approximates Bayesian inference by leveraging dropout layers, which are typically disabled during inference, as a source of stochasticity. By performing T stochastic forward passes with dropout active, an ensemble of predictions $\{\hat{\mathbf{y}}_t\}_{t=1}^T$ is obtained for each input, from which the following are computed:

$$\bar{\mathbf{y}} = \frac{1}{T} \sum_{t=1}^T \hat{\mathbf{y}}_t \quad (\text{predictive mean}) \quad (7)$$

$$\sigma^2(\mathbf{y}) = \frac{1}{T} \sum_{t=1}^T (\hat{\mathbf{y}}_t - \bar{\mathbf{y}})^2 \quad (\text{predictive variance}) \quad (8)$$

The 95% confidence interval at each spatial location is then $\bar{\mathbf{y}} \pm 1.96 \sigma(\mathbf{y})$.

Implementation.

Dropout layers are added to the Transformer attention blocks and feed-forward networks in AB-UPT. At inference time, $T = 30$ forward passes are performed per geometry, and the predictions are aggregated. The dropout rate is set to 0.1, matching the training configuration.

Loss function.

No modification is made to the training loss to account for data noise. Kendall and Gal [10] showed that augmenting the loss with a learned per-point log-variance term enables simultaneous estimation of aleatoric (data) and epistemic (model) uncertainty. However, this heteroscedastic formulation is designed for datasets with inherent measurement noise. In the present setting, the training data are deterministic CFD simulation outputs with no stochastic observation process; each geometry–field pair is a unique solution of the governing equations, not a noisy sample from a distribution. Under this assumption of zero aleatoric uncertainty, the standard MSE loss is retained and MC Dropout captures only the epistemic component.

Key metrics.

The primary outputs of MC Dropout are:

- **Predictive variance $\sigma^2(\mathbf{y})$:** Measures the spread of the stochastic ensemble at each spatial point. Physically, a high σ^2 on the vehicle surface indicates a region where the model’s learned parameters are sensitive to perturbation, i.e., the mapping from geometry to physical field is not well-constrained by the training data in that region. *Expected behavior:* σ should be on the same order of magnitude as the actual prediction error (MAE); $\sigma \ll \text{MAE}$ indicates overconfidence, while $\sigma \gg \text{MAE}$ indicates excessive conservatism.
- **95% confidence interval $\bar{\mathbf{y}} \pm 1.96 \sigma$:** Under a Gaussian approximation, 95% of ensemble predictions fall within this interval. Mathematically, this bounds the epistemic component of the prediction error. When the true value falls outside this interval, the model is overconfident at that location. *Expected behavior:* The width should be large enough to achieve the nominal coverage (see CI coverage below), but not so wide as to be uninformative.
- **Coefficient of variation $\text{CV} = \sigma/|\bar{\mathbf{y}}|$:** A dimensionless measure of relative uncertainty, enabling comparison across fields with different physical scales (e.g., pressure vs. velocity). *Expected behavior:* Domain-dependent; useful primarily for cross-field and cross-sample ranking rather than absolute thresholding.
- **CI coverage:** The fraction of true values that fall within the predicted 95% confidence interval, evaluated over a dataset. This is the primary diagnostic for

calibration quality. *Expected behavior*: A well-calibrated model should achieve coverage $\approx 95\%$ (matching the nominal CI level). Coverage significantly below 95% indicates *overconfidence*—the model’s uncertainty estimates are too narrow—while coverage significantly above 95% indicates *underconfidence*. CI coverage is the single most direct indicator of whether the model is overconfident.

- **$\bar{\sigma}$ /MAE ratio**: The ratio of the mean MC standard deviation to the mean absolute error of the point prediction. This provides a scale-free measure of how well the uncertainty magnitude matches the actual error magnitude. *Expected behavior*: $\bar{\sigma}/\text{MAE} \approx 1$ indicates that the model’s estimated uncertainty is commensurate with its true errors. $\bar{\sigma}/\text{MAE} \ll 1$ signals severe overconfidence (the model “knows” less than it claims), while $\bar{\sigma}/\text{MAE} \gg 1$ indicates excessive conservatism.

4.3 Complementary Usage

The two methods address different use cases and can be combined in a practical workflow:

1. **Screening (Mahalanobis distance)**: For a new geometry, compute D_M with a single forward pass. If D_M exceeds the training distribution threshold, flag the geometry as OOD and recommend running a full CFD simulation. For in-distribution geometries, D_M additionally serves as a continuous reliability indicator: lower values suggest higher prediction confidence.
2. **Detailed assessment (MC Dropout)**: For geometries that pass the OOD screen, optionally run MC Dropout inference to obtain spatially resolved uncertainty maps, identifying specific surface regions where the model is least confident. This is particularly useful for safety-critical applications or when prediction quality in specific regions is of interest.

This two-stage workflow is validated empirically in Sections 5.1 and 5.4.

5 Experiments

5.1 Mahalanobis Distance Results

5.1.1 Experimental Setup

To systematically evaluate OOD detection capability, five models were trained with progressively increasing numbers of vehicle types, starting from a single vehicle (Mid-size SUV, 102 samples) and adding one vehicle type at a time until all five types were included (511 samples). At each stage, 80% of the available samples were used for training and the remainder for validation and testing. The full breakdown is provided in Table 3.

For each model, Mahalanobis statistics ($\mu, \hat{\Sigma}^{-1}$) were computed from the training split, and D_M scores were evaluated on all 511 geometries.

5.1.2 Progressive Expansion Results

Table 3 shows overall Mahalanobis distance statistics as more vehicle types are added to training.

Table 3: Overall Mahalanobis distance statistics across 511 geometries for models trained with increasing numbers of vehicle types. The training reference row shows the in-distribution baseline.

Training Set	N_{train}	Train (ref)			All 511		
		mean	p95	max	mean	p95	max
1 veh (Midsize SUV)	91	7.52	8.00	8.46	34.87	52.98	68.04
2 veh (+Small Crossover)	164	8.00	9.17	9.91	16.46	25.23	27.85
3 veh (+Large SUV #1)	246	8.90	9.98	11.39	16.58	32.53	36.48
4 veh (+Large SUV #2)	326	9.12	10.45	12.60	13.72	31.18	36.28
5 veh (all)	408	9.42	10.75	13.57	9.96	11.61	14.47

Key observations:

- When all five vehicles are included in training, the overall D_M distribution tightly concentrates around the training mean (9.96 vs. 9.42), with a maximum of 14.47, indicating that all samples are in-distribution.
- With fewer vehicle types, the overall mean is inflated by OOD samples, and the gap between training max and overall max grows dramatically.

5.1.3 Per-Vehicle Analysis: OOD Separation

Table 4 shows per-vehicle D_M statistics for the 4-vehicle model (Midsize SUV + Small Crossover + Large SUV #1 + Large SUV #2), where Large SUV #3 is the unseen vehicle type.

Table 4: Per-vehicle Mahalanobis distance for the 4-vehicle model. Large SUV #3 is the OOD vehicle (not seen during training). Training reference: mean= 9.12, p95= 10.45.

Vehicle	N	D_M mean	D_M std	D_M p50	D_M p95	D_M max
Midsize SUV	102	9.46	0.85	9.33	11.19	11.78
Small Crossover	103	10.33	0.92	10.22	11.87	13.62
Large SUV #1	103	9.10	0.67	9.02	10.16	11.52
Large SUV #2	100	9.60	0.79	9.55	11.00	12.04
Large SUV #3 (OOD)	103	29.96	2.03	29.78	33.90	36.28

The separation is striking: Large SUV #3’s mean D_M (29.96) is approximately $3\times$ that of any training vehicle, with *zero overlap* between Large SUV #3’s minimum score

(26.32) and the maximum score of any training vehicle (13.62). This demonstrates that Mahalanobis distance provides a clear, unambiguous OOD signal for unseen vehicle types.

Crucially, this separation would not be detectable from conventional aerodynamic summary statistics. As shown in Figure 1, Large SUV #3’s C_d and frontal area distributions overlap substantially with those of the training vehicles; a simple C_d -area scatter plot would not flag Large SUV #3 as anomalous. The Mahalanobis distance, computed from the model’s learned 512-dimensional geometric feature space, captures shape-level differences (body curvature, panel geometry, underbody configuration) that are invisible to low-dimensional scalar summaries. This highlights the value of feature-space OOD detection: it identifies distributional novelty in the aspects of geometry that actually matter for flow prediction, rather than relying on global quantities that may be coincidentally similar.

5.1.4 Scaling Behavior

Figure 3 visualizes the effect of progressively expanding the training set on the D_M distribution. All panels share identical axis ranges (x: 5–70, y: 0–0.70 density) to enable direct visual comparison.

For the 1-vehicle model (Midsize SUV only, Figure 3a), all other vehicles have elevated D_M (overall mean = 34.87), with Large SUV #3 and Large SUV #2 forming a long tail extending past $D_M = 60$. Adding Small Crossover (Figure 3b) reduces the overall mean to 16.46, but Large SUV #1, Large SUV #3, and Large SUV #2 remain clearly separated. By the time all five vehicles are included (Figure 3e), the overall distribution collapses to a tight range (mean ≈ 10 , max ≈ 14.5), confirming that the model has learned a feature space that encompasses the full diversity of the training set.

5.1.5 Prediction Accuracy vs. Mahalanobis Distance

To examine how Mahalanobis distance relates to actual prediction quality, Figure 4 presents scatter plots of true vs. predicted total aerodynamic drag (F_x) for each progressive-expansion model, with points colored by their D_M score. In all four panels, the colormap is reversed so that higher Mahalanobis distance (darker color) corresponds to greater distributional novelty.

A consistent pattern emerges across all training-set sizes: samples with low D_M (lighter colors) cluster tightly along the identity line, indicating accurate predictions, while samples with high D_M (darker colors) tend to deviate further from the diagonal. This correlation is most pronounced for the 1-vehicle model (Figure 4a), where unseen vehicle types exhibit both the highest D_M values and the largest prediction errors. As the training set expands to include more vehicle types (Figures 4b to 4d), the overall D_M range narrows and predictions improve, but the positive association between Mahalanobis distance and prediction error persists.

This result provides direct empirical evidence that D_M is not merely an abstract distributional statistic but a practically informative indicator of prediction reliability: higher Mahalanobis distance signals greater risk of inaccurate force predictions.

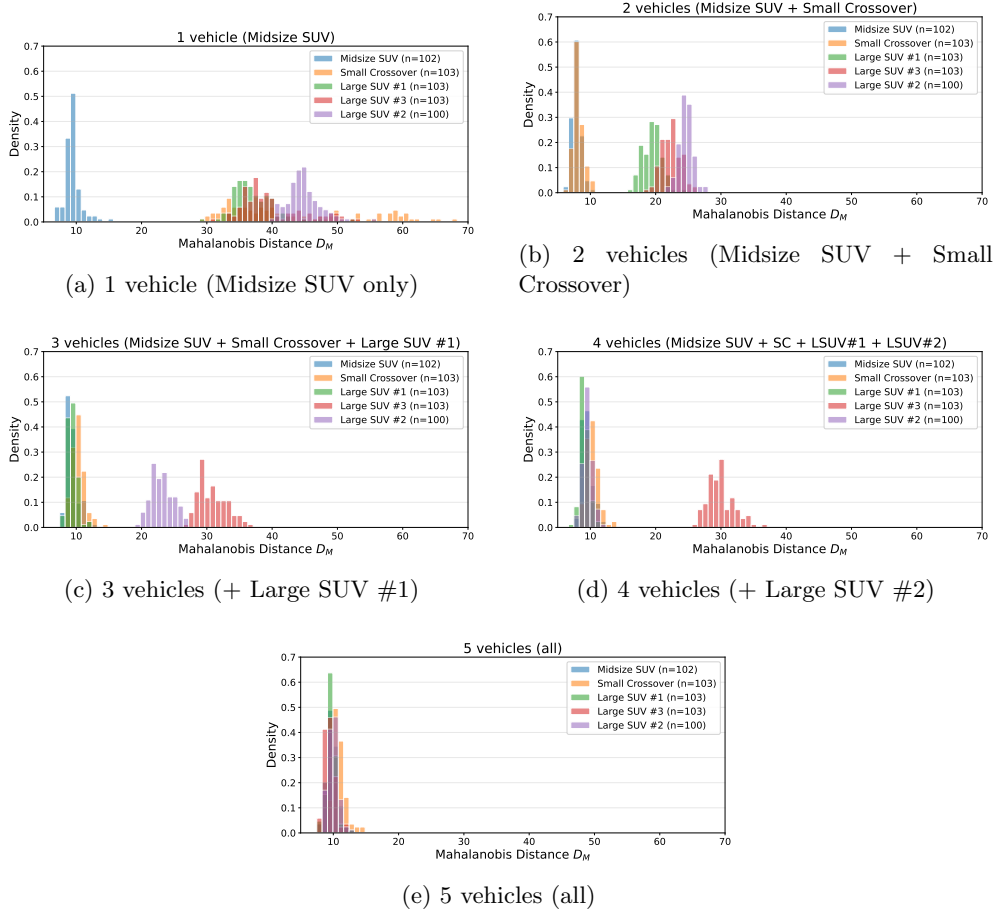
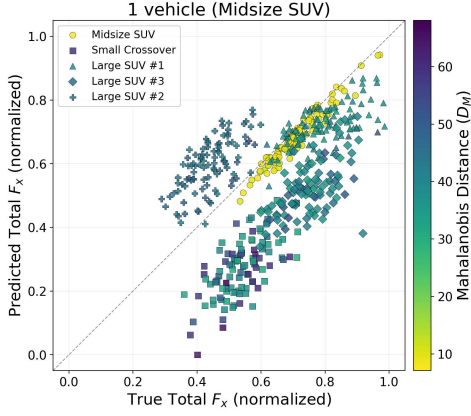


Fig. 3: Mahalanobis distance distributions by vehicle type as the training set is progressively expanded from 1 to 5 vehicle types. All 511 geometries are evaluated against each model’s training statistics. OOD vehicles (those not included in training) form distinct peaks at higher D_M values, clearly separated from in-distribution vehicles. When all 5 vehicles are included (e), the distributions collapse into a single tight cluster.

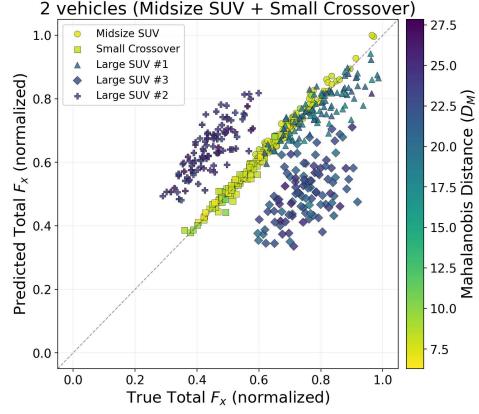
5.2 Discussion: Practical OOD Threshold

A natural threshold for OOD classification can be derived from the training distribution. Using the 4-vehicle model as an example:

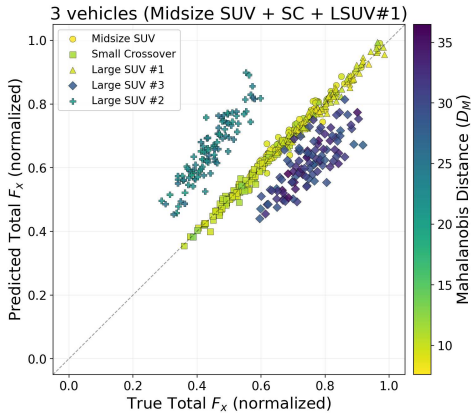
- Training D_M p95 = 10.45, max = 12.60
- All ID vehicles fall within $D_M < 14$ (with outliers at 13.62)
- Large SUV #3 (OOD) minimum = 26.32



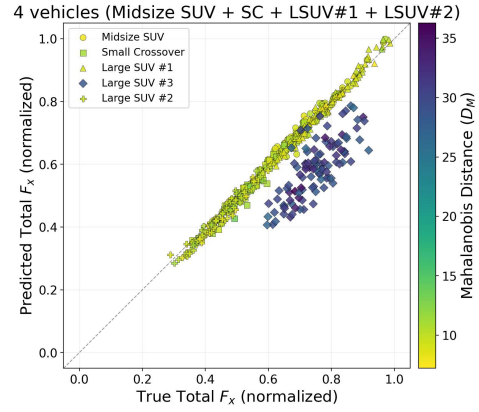
(a) 1 vehicle (Midsize SUV only)



(b) 2 vehicles (Midsize SUV + Small Crossover)



(c) 3 vehicles (Midsize SUV + Small Crossover + Large SUV #1)



(d) 4 vehicles (Midsize SUV + Small Crossover + Large SUV #1 + Large SUV #2)

Fig. 4: True vs. predicted total F_x (drag), min-max normalized to $[0, 1]$, colored by Mahalanobis distance D_M for progressive vehicle expansion. Darker points indicate higher D_M (greater distributional novelty). Samples with high D_M consistently deviate further from the identity line, demonstrating that Mahalanobis distance is a reliable predictor of force prediction accuracy.

A threshold of $D_M > 15$ would achieve perfect separation in this dataset, with a comfortable margin. In practice, the training p99 or max is recommended as a conservative threshold, with the understanding that the gap between max-ID and min-OD (12.60 $\rightarrow$ 26.32) provides substantial safety margin.

5.3 Limitations: Sample Size and Feature Dimension

An important caveat arises when the number of training samples N is much smaller than the feature dimension d . In a LoRA fine-tuning experiment where a pretrained 4-vehicle model was adapted to Large SUV #3 data (only $N = 20$ training samples with $d = 512$ features), the Mahalanobis distance exhibited pathological behavior:

- Training D_M mean collapsed to 4.53 (vs. expected $\sqrt{d} \approx 22.6$ under χ^2 reference)
- The sample covariance had condition number 2.33×10^{12}
- OOD samples reached $D_M > 1000$, rendering the scores meaningless

This occurs because 20 samples cannot span a 512-dimensional space: the rank-deficient covariance matrix, even after shrinkage ($\alpha = 0.12$), produces a precision matrix that dramatically overweights directions not represented in the training data. This limitation should be considered when applying Mahalanobis distance to few-shot learning scenarios.

For cross-model comparison (e.g., pretrained vs. LoRA-finetuned), a more appropriate approach would be to use a common reference distribution (e.g., the pretrained model’s statistics with $N = 326$) and compare the feature distributions produced by each model under that shared reference frame.

5.3.1 Implications for Industrial SciML

The $N > d$ requirement is not merely a theoretical inconvenience; it represents a fundamental constraint for industrial Scientific Machine Learning applications. In domains such as computational fluid dynamics, structural analysis, and thermal simulation, each training sample is the product of a high-fidelity simulation that may take hours to days of compute time. Acquiring hundreds or thousands of simulation samples is often prohibitively expensive.

Consider the practical numbers in this study: the largest training set contains $N = 408$ samples against a feature dimension of $d = 512$, yielding $N/d = 0.80$. While Ledoit–Wolf shrinkage produces a usable precision matrix (with shrinkage coefficient $\alpha \approx 0.05$), the resulting Mahalanobis distance is partially “Euclideanized”: the regularization toward $\mathbf{I}$ means that learned inter-feature correlations are partially discarded. For the LoRA scenario ($N/d = 0.04$), the method breaks down entirely.

This tension between feature dimensionality and sample availability is characteristic of industrial SciML:

- Modern neural operators such as AB-UPT use embedding dimensions of $d = 256$ – 1024 to capture complex physics, making d large by design.
- Simulation-generated datasets typically contain 10^1 – 10^3 samples, far fewer than the 10^4 – 10^6 samples common in computer vision or NLP where Mahalanobis-based OOD detection was originally developed [13].
- Transfer learning and few-shot adaptation (e.g., LoRA fine-tuning to a new vehicle type) further reduce N , exacerbating the problem.

Potential mitigations include: (1) dimensionality reduction of the feature space (e.g., PCA to the top $k \ll N$ components before computing D_M), (2) using a shared reference distribution from a pretrained model with larger N , or (3) alternative OOD scores that do not require full covariance estimation, such as k -nearest neighbor distances in feature space.

5.4 MC Dropout Results

MC Dropout inference was performed on a model trained with all five vehicle types (511 samples, surface-only mode) with dropout rate 0.1 and $T = 30$ stochastic forward passes per sample. Note that the surface-only model was deliberately chosen for this experiment due to the prohibitive computational cost of MC Dropout on the combined (surface + volume) model. Each combined-mode inference takes approximately 2 minutes per sample; performing $T = 30$ stochastic forward passes over all 511 samples would therefore require $30 \times 511 \times 2 = 30,660$ minutes ≈ 511 hours ≈ 21 days of continuous GPU time, which is impractical. In contrast, the surface-only model reduces per-sample inference time sufficiently to make MC Dropout feasible within a reasonable time budget.

For the same reason, MC Dropout was applied only to the single 5-vehicle model. Unlike the Mahalanobis distance experiments (Section 5.1), where progressive vehicle expansion from 1 to 5 types was systematically evaluated, a comparable set of MC Dropout experiments, training separate dropout-enabled models for each vehicle subset and running $T = 30$ inference passes on all 511 samples, was not conducted. Such an experiment campaign would have required training 5 additional models and performing $5 \times 30 \times 511$ stochastic forward passes, representing a prohibitive computational investment with limited expected additional insight.

5.4.1 Force Prediction Uncertainty

Table 5 summarizes the MC Dropout uncertainty metrics for integrated aerodynamic forces across the full dataset.

Table 5: MC Dropout uncertainty metrics for aerodynamic force predictions (511 samples, $T = 30$). $\bar{\sigma}$: mean MC standard deviation; CI width: mean 95% confidence interval width; CI coverage: fraction of true values within the 95% CI (nominal: 95%); MAE: mean absolute error of the MC mean prediction; $\bar{\sigma}/\text{MAE}$: ratio of estimated uncertainty to actual error (ideal: ≈ 1). CV ($= \sigma/|\bar{y}|$) is omitted from this table because the mean integrated side force $\bar{F}_y \approx 0$, making CV ill-defined; CV is more informative for surface fields (Section 4.2).

Force Component	$\bar{\sigma}$ [N]	Mean CI Width [N]	CI Coverage	MAE [N]	$\bar{\sigma}/\text{MAE}$
Total F_x (drag)	10.57	7.90	47.7%	5.00	2.11
Total F_y (side)	0.55	0.41	3.5%	6.23	0.09
Total F_z (lift)	5.02	3.75	8.6%	14.27	0.35

Overconfidence. The CI coverage is notably low ($< 50\%$ for all components), indicating that the model is *overconfident*: the MC Dropout confidence intervals are too narrow to capture the actual prediction errors. A well-calibrated 95% CI should contain the true value 95% of the time; the observed coverages of 47.7%, 3.5%, and 8.6% fall far short.

The degree of overconfidence varies dramatically across force components. For Total F_x , the mean MC standard deviation ($\bar{\sigma} = 10.57$ N) is roughly twice the mean absolute error (MAE = 5.00 N), yet CI coverage is only 47.7%, suggesting that the MC variance captures the right order of magnitude but the error distribution has heavier tails than assumed by Gaussian CIs. For Total F_z , the mismatch is more severe: $\bar{\sigma} = 5.02$ N versus MAE = 14.27 N, meaning the MC standard deviation is approximately $3\times$ *smaller* than the actual errors. For Total F_y (side force), the situation is extreme: $\bar{\sigma} = 0.55$ N versus MAE = 6.23 N, a factor of $11\times$ underestimate.

This systematic overconfidence is consistent with the known limitations of MC Dropout [14]: dropout variance primarily captures how sensitive the model’s predictions are to the random zeroing of hidden units, which is only a rough proxy for the true epistemic uncertainty. When prediction errors are dominated by systematic biases (e.g., the model consistently under- or over-predicts lift for certain vehicle geometries), dropout perturbations, which act as random noise, cannot produce variance large enough to cover these biases.

Per-vehicle uncertainty. Figure 5 shows the MC Dropout prediction for Total F_x on the Midsize SUV vehicle type as a representative example. Error bars denote $\pm 2\sigma$ (approximately 95% CI). While the predictions track the ground truth well ($R^2 = 0.899$ on training, 0.827 on validation, 0.793 on test), the error bars are visibly too narrow to envelop many of the off-diagonal points, particularly for the validation and test sets.

Table 6 provides per-vehicle CI coverage for all three force components. The coverage varies substantially across vehicles, from 35.9% (Large SUV #3, F_x) to 61.0% (Large SUV #2, F_x), but remains well below the nominal 95% for every combination. Validation-set coverage is consistently lower than overall coverage, confirming that the overconfidence is more pronounced on unseen samples.

Table 6: Per-vehicle MC Dropout CI coverage (%) for integrated forces ($T = 30$, nominal 95% CI). All values are well below the expected 95%, confirming systematic overconfidence.

Vehicle	Total F_x	Total F_y	Total F_z
Midsize SUV	48.0	4.9	15.7
Small Crossover	52.4	4.9	8.7
Large SUV #1	41.7	1.0	4.9
Large SUV #3	35.9	2.9	6.8
Large SUV #2	61.0	4.0	7.0
Overall	47.7	3.5	8.6

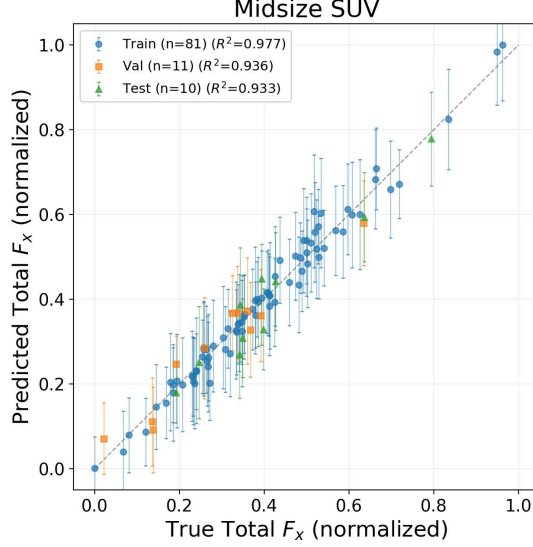


Fig. 5: MC Dropout predictions vs. ground truth for Total F_x on Midsize SUV (102 samples). Error bars show $\pm 2\sigma$ from $T = 30$ stochastic forward passes. Despite a strong correlation ($R^2 = 0.90$), many validation/test points fall outside the confidence intervals, illustrating overconfidence.

Qualitative value of MC Dropout. Despite the poor calibration, MC Dropout does provide useful *relative* uncertainty information. Figure 6 shows the distribution of MC standard deviations, mean absolute errors, and CI coverage across all 511 samples for each force component. While the absolute scale of σ is unreliable, the relative ranking, identifying which samples have higher versus lower uncertainty, can still guide practitioners toward cases that warrant closer inspection.

Spatially resolved uncertainty visualization. A key strength of MC Dropout over global OOD metrics such as the Mahalanobis distance is its ability to produce *spatially resolved* uncertainty maps. By computing the pointwise predictive variance $\sigma^2(\mathbf{x})$ across the vehicle surface, the MC ensemble yields a 3-D uncertainty field that can be directly overlaid on the geometry. This allows engineers to identify *where* on the surface the model is least confident—for example, regions of flow separation, sharp geometric features, or areas underrepresented in the training data—rather than receiving only a single scalar confidence measure per sample.

Figure 7 illustrates this for a representative Midsize SUV sample, comparing the pointwise MC standard deviation $\sigma(\mathbf{x})$ with the actual absolute prediction error $|p^{\text{true}} - \bar{p}^{\text{pred}}|$. The spatial correspondence is striking: regions of elevated MC uncertainty—such as the front fascia, wheel arches, and underbody separation zones—coincide with the regions where the model incurs the largest prediction errors. This confirms that, despite the poor *absolute* calibration of MC Dropout at the integrated-force level (Table 5), the *spatial* uncertainty signal is informative: MC Dropout

MC Dropout Uncertainty Distribution — Overall

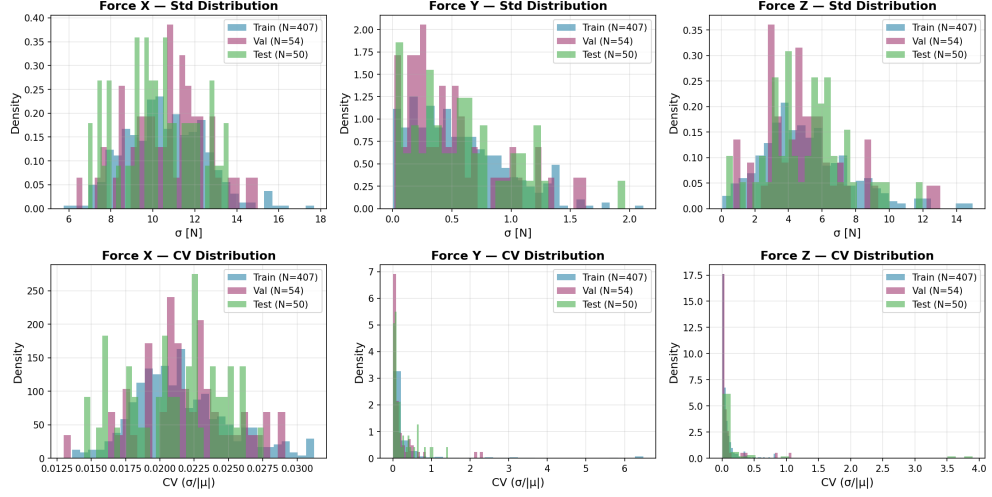


Fig. 6: MC Dropout uncertainty overview across all 511 samples. Left: distribution of MC standard deviations by force component. Center: mean absolute error distribution. Right: CI coverage distribution. The consistently low CI coverage confirms model-wide overconfidence.

correctly identifies *where* on the surface the model struggles, even if it cannot reliably quantify *how much* it will err.

5.4.2 Field Prediction Accuracy

Table 7 summarizes the deterministic prediction accuracy (using the MC mean) for surface-integrated forces.

Table 7: Deterministic prediction accuracy for surface force components (R^2 and NRMSE).

Force Component	R^2	NRMSE
Pressure F_x	0.975	15.7%
Pressure F_z	0.987	11.5%
Shear F_x	0.935	25.4%
Total F_x	0.973	16.3%
Total F_z	0.987	11.4%

The model achieves $R^2 > 0.97$ for pressure-dominated force components (drag, lift), confirming strong predictive capability for the primary quantities of interest.

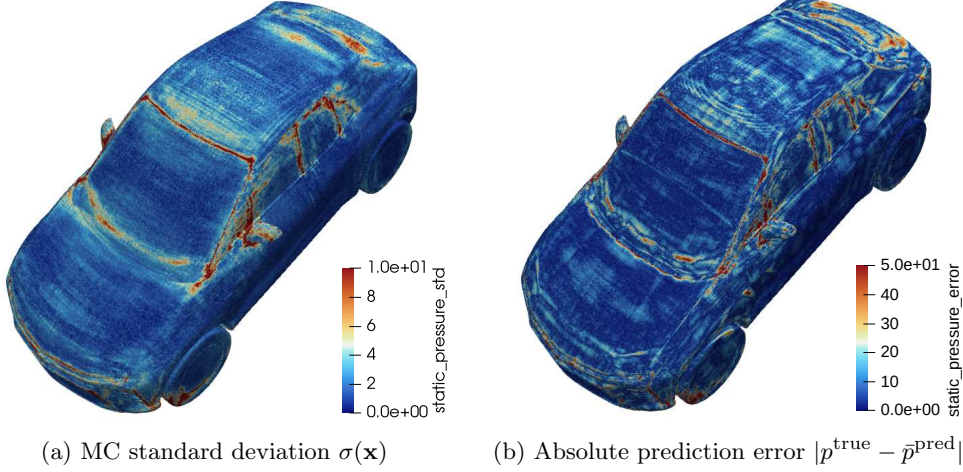


Fig. 7: Spatially resolved MC Dropout uncertainty vs. actual prediction error on a Midsize SUV surface. (a) Pointwise standard deviation from $T = 30$ stochastic forward passes. (b) Absolute error between the MC mean prediction and CFD ground truth. High-uncertainty regions in (a) spatially coincide with high-error regions in (b), demonstrating that MC Dropout provides a reliable spatial indicator of prediction quality despite its poor absolute calibration.

5.4.3 MC Dropout on OOD Geometries

While MC Dropout does not directly classify a geometry as OOD, its uncertainty estimates exhibit a characteristic signature when applied to unseen vehicle types. To investigate this, an additional 4-vehicle model (Midsize SUV + Small Crossover + Large SUV #1 + Large SUV #2) was trained with dropout rate 0.1, and MC Dropout inference with $T = 10$ stochastic forward passes was performed on all 511 geometries, including the OOD vehicle Large SUV #3.

Figure 8 compares the MC Dropout predictions for Total F_x on Midsize SUV (in-distribution) and Large SUV #3 (out-of-distribution). The reduced number of forward passes ($T = 10$ vs. $T = 30$ in Section 5.4) introduces additional sampling noise, but the relative behavior between ID and OOD samples remains clearly interpretable.

For the in-distribution Midsize SUV (Figure 8a), the MC mean predictions track the ground truth well ($R^2 = 0.979$ on training, 0.915 on validation) with a mean MC standard deviation of $\bar{\sigma} = 6.43$ N and RMSE = 9.8 N. The error bars, while still too narrow for full CI coverage, are visually consistent with the observed scatter.

For the OOD vehicle Large SUV #3 (Figure 8b), the picture changes dramatically. The predictions are systematically biased low (most points lie well below the identity line), with RMSE = 75.4 N—an order of magnitude worse than Midsize SUV. Crucially, the MC standard deviation also increases to $\bar{\sigma} = 8.60$ N, approximately 34% higher than Midsize SUV’s. This increase reflects the model’s internal sensitivity: when presented with a geometry that differs substantially from its training experience,

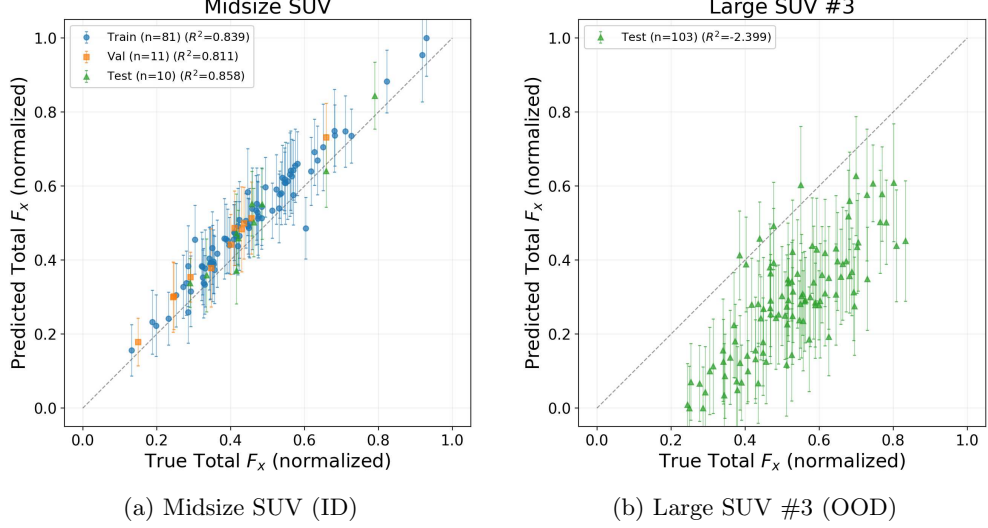


Fig. 8: MC Dropout predictions ($T = 10$) for Total F_x on a 4-vehicle model. (a) Midsize SUV (in-distribution) shows tight clustering around the identity line with moderate error bars. (b) Large SUV #3 (out-of-distribution) exhibits both large systematic errors (predictions biased low) and elevated MC standard deviation, with $R^2 = -4.55$ indicating predictions worse than a constant baseline.

dropout perturbations produce more diverse predictions, as the learned representations for OOD inputs are less robust to parameter noise.

However, the increase in σ (+34%) is vastly outpaced by the increase in actual error (+670%). The $\bar{\sigma}$ /RMSE ratio drops from 0.66 (Midsize SUV) to 0.11 (Large SUV #3), meaning that MC Dropout becomes *even more overconfident* on OOD data. The confidence intervals widen somewhat, but the prediction errors grow far faster, so the model remains confidently wrong.

MC Dropout as an indirect OOD indicator. Although MC Dropout cannot directly classify a geometry as ID or OOD, the elevated MC standard deviation on OOD samples suggests an indirect screening criterion: if the MC variance for a new geometry is substantially higher than the typical ID range (e.g., $\bar{\sigma} \gg 6-7$ N for Total F_x), this signals that the model’s learned features are unstable under dropout perturbations, which is a hallmark of OOD inputs. In practice, this complements the Mahalanobis distance approach: D_M provides a direct, computationally efficient OOD flag, while elevated MC variance serves as a secondary confirmation when MC Dropout inference is performed for other purposes.

5.4.4 Key Findings from MC Dropout

The MC Dropout experiments reveal several important lessons for UQ in transformer-based SciML models:

1. **Systematic overconfidence.** The 95% CI coverage ranges from 3.5% (F_y) to 47.7% (F_x), far below the nominal 95%. The model’s dropout-induced variance consistently underestimates the true prediction error, particularly for force components where systematic biases dominate (side force, lift).
2. **Component-dependent calibration.** Overconfidence severity varies by $> 10\times$ across force components: the MC standard deviation-to-MAE ratio is ~ 2.1 for F_x (moderate) but ~ 0.09 for F_y (extreme). This suggests that dropout perturbations interact differently with the model’s internal representations for different output quantities.
3. **Relative uncertainty is informative.** Although the absolute scale of MC Dropout variance is poorly calibrated, it still provides a meaningful relative ranking of prediction confidence across samples. For Total F_x , the correlation between MC variance and actual error is positive, enabling identification of higher-risk predictions, albeit not at the level needed for reliable confidence intervals.
4. **The calibration problem is fundamental, not parametric.** While increasing the dropout rate might improve coverage, it would also degrade prediction accuracy (the MC mean would become a worse point estimator). To verify this empirically, we trained an additional model with the dropout rate increased from 0.1 to 0.3 and repeated the MC Dropout evaluation; the CI coverage remained far below 95% across all force components, confirming that the miscalibration is not resolved by simply widening the dropout perturbation (detailed results omitted for brevity). The core issue is that dropout perturbations are a crude approximation to posterior sampling: they act as additive noise in the hidden units, which cannot capture the structured, geometry-dependent biases that dominate prediction errors in this application.
5. **Spatial uncertainty is informative despite poor absolute calibration.** Although MC Dropout fails to produce reliable confidence intervals at the integrated-force level, its *spatial* uncertainty signal is empirically valid. Comparison of the pointwise MC standard deviation $\sigma(\mathbf{x})$ with the actual prediction error on the vehicle surface (Figure 7) reveals strong spatial correspondence: regions where the model reports high dropout sensitivity coincide with regions of large prediction error. This makes MC Dropout a useful diagnostic for identifying problematic surface regions, even when its quantitative uncertainty estimates cannot be trusted.

5.5 Discussion: MC Dropout and Transformer Architectures

While MC Dropout is a widely adopted UQ method due to its simplicity, requiring only that dropout be enabled at inference time, its computational overhead scales linearly with the number of stochastic forward passes T . For transformer-based architectures such as AB-UPT, where a single forward pass already involves substantial computation (multi-head self-attention and cross-attention over large point sets), this $T\times$ multiplier becomes a decisive practical bottleneck.

As demonstrated in Section 5.4, even with the lighter surface-only model, $T = 30$ passes over 511 samples required significant GPU resources. The combined (surface + volume) model would have required approximately 21 days of continuous computation

for the same analysis, rendering the approach infeasible for routine deployment. This stands in contrast to Mahalanobis distance, which requires only a single forward pass per sample regardless of model complexity.

More broadly, this limitation suggests that MC Dropout is fundamentally ill-suited as a UQ method for large-scale transformer architectures in SciML applications. The inference cost of transformers, already a concern for deployment, is directly amplified by the $T\times$ factor, making it impractical for high-throughput screening or real-time uncertainty assessment. Alternative UQ methods that do not require repeated stochastic inference, such as deterministic uncertainty estimation, spectral-normalized neural Gaussian processes, or the Mahalanobis distance approach presented in this work, are more appropriate for transformer-based surrogate models.

5.6 Method Comparison

Table 8 synthesizes the practical characteristics of the two reliability assessment methods evaluated in this study.

Table 8: Comparison of Mahalanobis distance and MC Dropout for reliability assessment of neural CFD surrogates.

Criterion	Mahalanobis Distance	MC Dropout
Computational cost	1 forward pass + $O(d^2)$	$T\times$ forward pass ($T = 30$)
Output granularity	Scalar per geometry	Pointwise field
OOD detection	Direct (threshold-based)	Indirect (elevated σ)
Spatial localization	No	Yes
Calibration quality	N/A (score, not CI)	Poor (CI coverage 3.5–47.7%)
Error correlation	Strong	Moderate (relative ranking)
Sensitivity to N/d	High (requires $N/d \gtrsim 0.5$)	None

The two methods are complementary rather than competitive. Mahalanobis distance excels as a fast, per-geometry screening tool: a single forward pass produces a scalar score that reliably distinguishes OOD geometries and correlates with downstream prediction error. MC Dropout, despite its poor absolute calibration and high computational cost, provides spatially resolved diagnostics that no scalar metric can offer—identifying *where* on the vehicle surface the model is least confident. The practical recommendation is to deploy Mahalanobis distance as the primary reliability gate due to its negligible marginal cost, and reserve MC Dropout for targeted investigations of high-stakes predictions that pass the OOD screen.

6 Discussion: Deployment Considerations

6.1 Pre-inference OOD Detection

SciML inference is computationally intensive, and ideally one should determine whether a trained model is applicable to the geometry of interest before performing

inference. A lightweight OOD detection step before actual inference is recommended, so that precious computing resources can be saved.

Any feature-space distance metric could serve this role; in this study, the Mahalanobis distance is evaluated as a representative approach. It requires only a single forward pass to extract features, followed by a lightweight $O(d^2)$ matrix–vector product, making it orders of magnitude faster than repeated stochastic inference. It directly answers the OOD question—geometries whose D_M significantly exceeds the training distribution’s 95th percentile are flagged as out-of-distribution—and it does so in the model’s learned feature space, capturing geometric similarity as perceived by the model rather than relying on hand-crafted shape descriptors. Beyond binary classification, D_M also serves as a continuous indicator of prediction reliability: samples with higher Mahalanobis distance consistently exhibit larger force prediction deviations (Figure 4), making it informative not only for flagging OOD inputs but also for ranking the trustworthiness of in-distribution predictions.

These strengths come with limitations. Mahalanobis distance does not provide prediction-level confidence intervals or spatially resolved uncertainty; it produces a single scalar per geometry. The method assumes an approximately Gaussian feature distribution, and heavily non-Gaussian features may reduce the reliability of the distance as an OOD score. Most critically for industrial SciML applications, the method is sensitive to the ratio of training samples to feature dimension: when $N \ll d$, the precision matrix estimation degrades even with Ledoit–Wolf shrinkage, as demonstrated in the LoRA fine-tuning scenario (Section 5.3).

6.2 Post-inference Confidence Interval

Once a new geometry passes the OOD detection, it undergoes inference and the end-user is given the prediction. However, passing the OOD screen does not guarantee accuracy of such prediction. There is still uncertainty in the predicted values, which can impact any engineering decisions based upon them. A critical component here is the confidence interval.

Various methods can provide such confidence estimates, including deep ensembles, evidential networks, and stochastic inference approaches. In this study, MC Dropout is evaluated as a representative method due to its simplicity and compatibility with any pretrained model. Its distinguishing strength is its ability to produce spatially resolved uncertainty maps. Regions of high predictive variance on the vehicle surface indicate where the model is least confident, offering engineers actionable guidance for identifying problematic areas. This spatial signal is not merely theoretical: as shown in Figure 7, regions of elevated MC standard deviation coincide with regions of large prediction error, confirming that MC Dropout correctly identifies *where* the model struggles even when its absolute calibration is poor. Furthermore, confidence intervals can be computed for any output quantity—surface pressure, integrated forces, or velocity fields—providing a per-point measure of prediction spread.

However, these capabilities come at substantial computational cost. The inference time scales linearly with the number of stochastic forward passes T : with $T = 30$, MC Dropout requires 30× the computation of a single deterministic prediction, which becomes prohibitive for large transformer architectures as discussed in Section 5.5.

Additionally, MC Dropout does not directly answer the binary ID/OOD question for a new geometry. That said, elevated MC variance on OOD samples (Section 5.4.3) suggests that anomalously high σ relative to known in-distribution baselines can serve as an indirect OOD indicator, complementing the Mahalanobis distance approach.

7 Conclusion

This study assessed the prediction reliability of a transformer-based neural surrogate for automotive external aerodynamics, proposing a two-stage uncertainty quantification pipeline that brackets the inference process: pre-inference OOD detection to screen novel geometries, and post-inference confidence estimation to quantify spatial prediction uncertainty. The pipeline was validated using Mahalanobis distance and MC Dropout as representative implementations across 511 design variants of five vehicle types. The work is motivated by the observation that, as SciML surrogate accuracy on public benchmarks approaches saturation, the critical barrier to industrial adoption is no longer predictive performance but deployment trustworthiness: a trained model will never refuse to predict, and without explicit safeguards, silent extrapolation failures pose a direct risk to engineering decisions.

7.1 Key Findings

1. **Mahalanobis distance provides reliable OOD detection.** Using the geometry encoder’s learned feature space, Mahalanobis distance cleanly separates unseen vehicle types from in-distribution data. In the 4-vehicle experiment, the OOD vehicle (Large SUV #3) exhibited a mean D_M of 29.96 versus ~ 9 –10 for training vehicles, with zero overlap between distributions. Notably, Large SUV #3 is not an outlier in terms of conventional aerodynamic summary statistics (C_d , frontal area), yet the 512-dimensional feature space captures shape-level differences invisible to such scalar metrics. This provides a practical, deployable mechanism for flagging novel geometries.
2. **Progressive vehicle expansion validates the approach.** As training diversity increases from 1 to 5 vehicle types, the D_M distribution progressively tightens, confirming that the model’s feature space faithfully reflects geometric similarity. The transition from 4 vehicles (max $D_M = 36.28$) to 5 vehicles (max $D_M = 14.47$) quantitatively demonstrates the inclusion of previously OOD geometries into the learned distribution.
3. **MC Dropout is systematically overconfident.** MC Dropout ($T = 30$ passes) produces spatially resolved uncertainty maps, but the confidence intervals are severely miscalibrated: CI coverage ranges from 3.5% (F_y) to 47.7% (F_x), far below the nominal 95%. The overconfidence is particularly extreme for force components where systematic biases dominate (F_y : MC σ /MAE = 0.09). This is a fundamental limitation of dropout-based uncertainty in transformer architectures, not merely a hyperparameter issue. The $T \times$ computational overhead further limits practical applicability.
4. **MC Dropout provides indirect OOD signals.** When applied to OOD geometries, MC Dropout exhibits elevated standard deviations ($\bar{\sigma} = 8.60$ N for OOD

Large SUV #3 vs. 6.43 N for ID Midsize SUV), reflecting increased sensitivity of learned representations to dropout perturbations. However, the increase in σ (+34%) is vastly outpaced by the increase in actual prediction error (+670%), making the model even more overconfident on OOD data. Nonetheless, an anomalously high MC variance relative to known ID baselines can serve as a secondary OOD indicator, complementing Mahalanobis distance.

5. **Sample size matters for Mahalanobis distance.** When the training set is very small ($N = 20$) relative to the feature dimension ($d = 512$), the covariance estimation degrades severely, producing unreliable distance scores. A minimum N/d ratio of approximately 0.5–1.0 is recommended for stable Mahalanobis distance computation.

7.2 Two-Stage Uncertainty Quantification

For production deployment of SciML aerodynamic surrogates, a two-stage uncertainty quantification pipeline is recommended:

- **Pre-inference OOD detection (mandatory).** Mahalanobis distance screening should be performed on every geometry before inference. Its single-forward-pass cost and clear separation characteristics make it suitable for routine deployment with negligible overhead. A threshold based on the training distribution’s 99th percentile provides a robust decision boundary. Geometries exceeding this threshold should be flagged or rejected before committing computing resources to inference.
- **Post-inference confidence interval (optional, for critical decisions).** When a geometry passes the OOD screen and the prediction will inform critical engineering decisions, MC Dropout can be invoked to provide spatially resolved confidence intervals. This step is optional due to its substantial computational overhead ($T \times$ the inference cost), and should be reserved for high-stakes predictions where quantified uncertainty is required to support design choices.
- **Monitor the N/d ratio.** When fine-tuning or retraining with limited data, consider dimensionality reduction of the feature space before computing Mahalanobis statistics, or use shared reference statistics from a larger pretrained model.

7.3 Future Work

Several directions warrant further investigation:

- Calibration of MC Dropout rates to improve CI coverage, or alternative post-hoc recalibration methods (e.g., temperature scaling, conformal prediction) to correct the systematic overconfidence without retraining.
- Extension of the D_M –error correlation analysis (Section 5.1.5) toward a quantitative error bound: the present study demonstrates a qualitative positive association between Mahalanobis distance and prediction error, but a formal regression model mapping D_M to expected error magnitude would enable practitioners to attach numerical confidence to individual predictions.

- **Solution to OOD inputs:** OOD detection is a valuable tool for determining whether a given trained model is suitable for a new geometry of interest. However, a natural follow-up question arises—what should happen when a geometry is identified as OOD? Building an entirely new training set is often impractical. Transfer learning, particularly through low-rank adaptation (LoRA), offers a viable path to resolving the OOD problem with low-overhead fine-tuning of an existing model to accommodate the novel geometry. The authors are actively investigating this approach to provide a comprehensive pipeline for industrial deployment.

7.4 Broader Perspective

The contribution of this work is not algorithmic: neither MC Dropout nor Mahalanobis distance is new, and neither is claimed to be the only feasible choice for its respective role. Rather, the value lies in articulating and validating the two-stage pipeline concept itself—the idea that SciML inference should be bracketed by pre-inference OOD detection and post-inference confidence estimation—and in demonstrating that such a pipeline can be assembled from existing, off-the-shelf UQ components under realistic industrial constraints: limited simulation budgets, proprietary geometries, and transformer-scale models with high-dimensional feature spaces where $N < d$. As SciML surrogates move from benchmark competitions toward production engineering tools, uncertainty quantification is not an optional add-on but an indispensable element of the deployment pipeline. The authors hope that the empirical findings and practical guidelines presented here can serve as a reference for other industrial SciML applications facing similar challenges. Although the proposed pipeline is demonstrated on automotive external aerodynamics, its structure is not domain-specific: any SciML surrogate that maps geometry to physical fields faces the same risks of silent extrapolation and uncalibrated confidence. The two-stage bracketing concept—pre-inference OOD screening followed by post-inference confidence estimation—is directly applicable to other simulation domains, including crashworthiness, thermal management, and structural analysis, wherever data-driven surrogates are deployed to support engineering decisions.

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